

Groningen region, Netherlands

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Recognition

- Harmonic Rising Mathematician Award
- Chess.com Member of the Year 2021
- Caltech Mathathon 2026, accepted participant

Languages

Dutch, English

Tom de Groot

AI engineer | System and algorithm design, math and ML research

Profile

I work as an agentic engineer on full stack design, algorithms, and mathematical proofs and formalizations. At SoundImports I am the project owner of Acqui, our in-house purchasing and data analysis tool. Outside work, I co-authored a set theory preprint and orchestrated its full formalization in Lean. I like creating and solving competitive programming puzzles. Besides that, I play Magic: The Gathering and squash, go bouldering, and like hiking.

Experience

AI engineer | SoundImports — Current

- Project owner of Acqui, SoundImports' in-house purchasing and data analysis tool.
- Build the application end to end: algorithms, AWS hosting, data lake design, and UI/UX.

Research and formalization

Vopěnka's Principle without Choice: Preservation under Symmetric Extensions — September 2026

Tom de Groot and Wojciech Aleksander Wołoszyn | Preprint, arXiv:2609.06856

- Co-authored our paper using initial AI-led proofs, rewriting and adding substantially to them.
- Led the orchestration of a 300,000-line Lean proof over 5 days, repairing and auditing upstream papers and arguments along the way.

First Proof, Problem 9 — February 2026

- Built a persistent pipeline for a Lean-verified solution in 3 days and 2,500 lines of Lean. Daniel Litt identified it as the only credible formalized solution he knew of.

Open-source formalization — 2026

- Prill's problem on Daniel Litt's problem list: solved in all genera with GPT-5.6.
- New Foundations: 17,174-line Lean proof that NF refutes the weak partition principle.
- PrimeNumberTheoremAnd: merged Lean proofs built with Aristotle and AI agents.

Algorithmic and systems engineering

HeXO — 2026

- Built and trained a self-play MCTS engine with custom GPU kernels written in Triton, and designed a novel graph-search scheduling algorithm for it.

MoleculeResolver — 2026

- Refactored the system design and improved its RDKit stereochemistry scoring.

Bazaar_ML, ParetoProof, Alternaut, ProjectEulerBot

- ML training environment for The Bazaar, plus other systems projects and contributions.

Competitive programming

- Create and solve competitive programming puzzles, including five kata on Codewars.

Education

BSc Chemistry | University of Groningen — 2021 to 2025

Thesis: Evaluating and Optimizing NMR Prediction with Graph Neural Networks

- Worked on graph neural network methods for NMR prediction in the Pollice Group. Created clean datasets and ran hyperparameter optimization with Optuna on HPC using Slurm.